seqCNA Methods

seqCNA: copy number changes from sequencing data

seqCNA estimates copy number changes in a tumor from the ratio of its read coverage to that of a normal sample. It uses seqDNAcopy, V. Seshan's package of the read counting, GC correction and segmentation steps, built on DNAcopy, with mouse support added. The values below are the ones the pipeline uses.

Read counts

Fragments are counted in the BAM files of the tumor and the normal. A fragment is counted once, from the first read of the pair, if the read passes quality control, is not marked as a duplicate, is not a secondary alignment, has a mapped mate and a mapping quality of at least 20, and the fragment length (insert size) is 75 to 750 bp. The mapping quality filter is not applied on chromosome X. Each fragment is counted at its midpoint in 100 bp bins along the genome. Fragments whose midpoint falls in a blacklist of problematic regions are removed (436 regions, 11.7 Mb, in the human genome; there is no blacklist for mouse). The autosomes and chromosome X are counted; chromosome Y and the mitochondrial genome are not.

GC correction

The tumor counts are corrected for GC bias relative to the normal. Each 100 bp bin is given the GC content of the 1 kb window centered on it (pctGCdata). The tumor and normal counts are summed over the bins of each GC value, and the log2 ratio of tumor to normal is computed for each GC value, after scaling the tumor by the ratio of the total normal to the total tumor counts. A loess curve (span 0.25) is fitted to these log2 ratios against GC content, weighted by log2 of the sum of the tumor and normal counts at each GC value. The tumor count of each bin is then multiplied by 2^-f, where f is the fitted value at the GC content of the bin.

Bin size

For targeted sequencing (exome, IMPACT) the 100 bp bins are segmented as they are. For shallow whole-genome sequencing they are merged into larger bins, with the size chosen for each pair so that a bin at the lower quartile of the normal counts holds about 100 normal fragments: the normal

counts are summed in 50 kb bins (500 kb if the lower quartile is below 1), and the bin size is 100 x 50 kb divided by their lower quartile, rounded down to a multiple of 100 bp.

Segmentation

Bins with 35 or fewer normal fragments are removed; in targeted data this keeps the bins in well-covered capture targets. The log2 ratio of each remaining bin is log2(s x tumor + 1) - log2(normal + 1), where s is the ratio of the total normal to the total tumor counts of these bins. The log2 ratios are segmented by circular binary segmentation (CBS, DNAcopy::segment [Olshen 2004, Venkatraman 2007]) with each bin weighted by log2(normal + 1 - 35), the default significance level (alpha = 0.01), and undo.splits = "sdundo" with undo.SD = 2: a change point is removed if the means on its two sides differ by less than two standard deviations. CBS computes its p-values by permutation; the random seed is fixed (42), so the same counts always give the same segments.

Segment clustering and the baseline

Segments at nearly the same level are merged into clusters, which removes small differences in level between segments (adapted from FACETS [Shen 2016]). The segment means are sorted, and the two adjacent levels that are closest are merged as long as they differ by less than 0.04; the level of a merged cluster is the mean of the log2 ratios of its bins, weighted as in the segmentation. Each segment then takes the level of its cluster. The cluster with the smallest abs(level)/sqrt(number of bins), a level near 0 with many bins, is the baseline (the "diploid" cluster in the code) and is set to 0; all levels are shifted by the same amount, and the log2 ratios of the bins move with their segment.

seqCNA does not estimate purity or ploidy. The log2 ratios are relative to the baseline and are not converted to numbers of copies; the baseline is not known to be two copies.

Choice of the normal

Each tumor is compared with every normal in the normal set, which are normal samples processed and sequenced in the same way as the tumors; a sample that is in both sets is not compared with itself. For each pair the root mean square of the bin log2 ratios after clustering, sqrt(mean(log2R^2)), is computed over all bins, and the pair with the smallest value is kept: the normal whose coverage pattern is closest to that of the tumor. Only that pair is reported.

Significant segments

As in [Cheng 2015], the mean and standard deviation of the bin log2 ratios in the baseline cluster are computed. For each segment a two-sided normal test compares its level with that mean, using the standard deviation of the bins (not divided by the number of bins of the segment) as the scale, so the p-value does not depend on the length of the segment. The p-values of all the segments of a tumor, chromosome X included, are adjusted for multiple testing by the Benjamini-Hochberg false discovery rate (FDR) [Benjamini 1995].

Gene-level calls

Genes are represented by the exons of one canonical transcript per gene. For the human exome (GRCh37/b37) these are Ensembl release 75 transcripts, with the canonical transcript of each gene from the isoform lists of vcf2maf v1.6.12 (the MSKCC and UniProt overrides); nine genes have two transcripts in these lists, and both are used. For the Twist mouse exome (GRCm38/mm10) they are Ensembl release 83 transcripts, with the transcript that overlaps the most capture targets as the canonical one.

A gene is in a segment if one of its exons overlaps the segment. In each tumor a gene is given one segment: of the segments it overlaps, the one with the lowest log2 ratio, whether significant or not. The gene is called if that segment has FDR < 0.05 and an absolute log2 ratio greater than 1, a coverage ratio more than twice or less than half that of the baseline. This threshold is on the log2 ratio only; smaller changes are not called. Because the lowest segment is chosen first, a gene partly in a deletion is called as a deletion, and a gene partly in a gain is not called if the rest of it is in a segment with a lower log2 ratio.

Calls are made on the autosomes only. A call on chromosome X would need a sex-matched normal and the known sex of the sample, which are rarely available. Chromosome X is still counted, segmented and plotted.

Software

seqCNA 6.0.0 with R 4.5.1, DNAcopy 1.84.0, seqDNAcopy 0.0.4, pctGCdata 0.3.0, Rsamtools 2.26.0 and data.table 1.18.6.1. DNAcopy 1.75.5 fixed the weights used in weighted segmentation (earlier releases segmented every chromosome after the first with the weights of the first chromosome), so the results differ from those of seqCNA 5.x, which ran with an earlier DNAcopy.

References

- Benjamini, Y, Hochberg, Y; J R Stat Soc B 1995; 57(1): 289-300.
- Cheng, DT, et al; J Mol Diagn 2015 May; 17(3): 251-264.
- Olshen, AB, et al; Biostatistics 2004 Oct; 5(4): 557-572.
- Shen, R, Seshan, VE; Nucleic Acids Res 2016; 44(16): e131.
- Venkatraman, ES, Olshen, AB; Bioinformatics 2007; 23(6): 657-663.

Source code

- https://github.com/soccin/seqCNA
- https://github.com/soccin/seqDNAcopy